

Practical No. 3

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Aim : Gather Application Specific Requirements for Assimilate into RE (Requirement's Engineering) Model

1. Aim

To gather and document application-specific requirements for a Smart Education Platform and classify them into different types of software requirements as part of the Requirement Engineering process.

2. Objectives

- Understand the concept of Requirement Engineering.
- Identify the stakeholders of a Smart Education Platform.
- Gather application-specific requirements.
- Differentiate between functional and non-functional requirements.
- Identify user and system requirements.
- Organize the gathered requirements for incorporation into an RE Model.

3. Problem Statement

Smart Education Platform

An educational platform is required to support smart education, a concept that describes learning in the digital age. The platform should enable learners to learn more efficiently, effectively, flexibly, and comfortably through digital learning resources, personalized learning support, assessments, progress tracking, and communication. Teachers should be able to create and manage courses, learning materials, assignments, quizzes, and student progress. Administrators should manage users, courses, platform settings, and reports. The system should provide a simple, secure, accessible, and reliable learning environment.

4. Requirement Engineering

Requirement Engineering (RE) is the systematic process of eliciting, analyzing, documenting, validating, and managing the requirements of a software system.

Major RE activities:

- Inception / Requirement Elicitation
- Requirement Analysis
- Requirement Specification
- Requirement Validation
- Requirement Management

5. Stakeholder Identification

Sr. No.	Stakeholder	Requirement / Interest
1	Student / Learner	Access courses, study materials, assessments, personalized learning, and track progress
2	Teacher / Instructor	Create courses, upload content, conduct assessments, and monitor learners
3	Administrator	Manage users, courses, content, platform settings, and reports
4	Parent / Guardian	Monitor learner progress and academic activity where applicable
5	Content Provider	Provide or manage digital educational resources
6	System Administrator	Maintain platform, security, backups, and technical operations

6. Requirement Gathering / Elicitation

Requirements can be gathered using interviews, questionnaires, observation, meetings, surveys, brainstorming, document analysis, and prototyping.

Example interview questions:

Question	Sample Answer
What should a learner be able to do?	Register, log in, browse courses, access learning materials, attempt quizzes, submit assignments, and track learning progress.
How should teachers manage learning content?	Teachers should be able to create courses and upload or organize videos, notes, presentations, links, assignments, and quizzes.
Should the platform support personalized learning?	Yes. The platform should use learner activity and performance to recommend suitable learning resources or topics.
Should learners be able to learn flexibly?	Yes. Learners should be able to access available learning content at a convenient time and continue from their previous progress.

7. Functional Requirements

Functional requirements describe what the system should do.

ID	Functional Requirement
FR01	The system shall allow learners to register an account.
FR02	The system shall allow registered users to log in securely.
FR03	The system shall display available courses and learning resources.
FR04	The system shall allow learners to search and filter courses or topics.
FR05	The system shall allow learners to enroll in available courses.
FR06	The system shall allow learners to access digital learning materials.
FR07	The system shall record learner progress for courses and learning activities.
FR08	The system shall provide quizzes and assessments.
FR09	The system shall evaluate objective assessments and display results.
FR10	The system shall allow learners to submit assignments.
FR11	The system shall provide personalized learning recommendations based on learner activity or performance.
FR12	The system shall allow teachers to create and manage courses and learning content.
FR13	The system shall allow teachers to create quizzes, assignments, and assessments.
FR14	The system shall allow teachers to view learner performance and progress.
FR15	The administrator shall be able to manage users, courses, content, and platform settings.

8. Non-Functional Requirements

Non-functional requirements describe how well the system should perform.

ID	Non-Functional Requirement
NFR01	The system should provide a user-friendly and accessible interface.
NFR02	The system should respond to normal user requests within a few seconds.
NFR03	User passwords and personal information should be stored securely.
NFR04	The platform should be available whenever learners and teachers need to access learning resources.
NFR05	The system should support multiple learners and teachers simultaneously.
NFR06	The system should maintain learner, assessment, and course data securely.
NFR07	The platform should be compatible with commonly used web browsers and mobile devices.
NFR08	The system should be scalable to support increasing users, courses, and learning content.
NFR09	The platform should provide reliable backup and recovery of important data.

9. User Requirements

Learner: create an account, log in, search courses, enroll in courses, access learning materials, attempt quizzes, submit assignments, receive recommendations, and view learning progress.

Teacher / Instructor: create and manage courses, upload learning materials, create assessments, evaluate assignments, and monitor learner performance.

Administrator: manage users, courses, content, permissions, reports, and platform settings.

Parent / Guardian: where enabled, view learner progress and academic activity.

10. System Requirements

Login: The system shall authenticate the user using registered credentials and shall allow access only when valid credentials are provided.

Course Enrollment: The system shall allow a learner to select an available course and enroll in it. After enrollment, the course shall appear in the learner's course list.

Progress Tracking: The system shall record completed learning activities, quiz results, assignment status, and course progress for each learner.

Personalized Recommendation: The system shall analyze available learner activity or performance data and provide relevant learning resources or topics.

11. Requirement Classification

Requirement Category	Example
Functional	Learner can enroll in a course.
Non-Functional	System should respond quickly.
User Requirement	Learner can view course progress.
System Requirement	System shall store completed learning activities in the database.
Business Requirement	Educational institution wants to provide flexible and smart digital learning.
Security Requirement	Learner and teacher data must be protected.

12. Requirement Specification

Req. ID	Requirement	Type	Priority	Stakeholder
FR01	User registration	Functional	High	Learner
FR02	User login	Functional	High	All Users
FR03	View/search courses	Functional	High	Learner
FR04	Course enrollment	Functional	High	Learner
FR05	Access learning materials	Functional	High	Learner
FR06	Track learning progress	Functional	High	Learner
FR07	Quizzes and assessments	Functional	High	Learner / Teacher
FR08	Personalized recommendations	Functional	Medium	Learner
FR09	Manage courses and content	Functional	High	Teacher
FR10	Manage users and platform	Functional	High	Administrator
NFR01	User-friendly interface	Non-functional	High	All Users
NFR02	Secure learner data	Non-functional	High	All Users
NFR03	Fast response	Non-functional	High	Learner
NFR04	High availability	Non-functional	High	All Users

13. Requirement Validation

Requirements should be checked for: Correctness, Completeness, Consistency, Feasibility, Unambiguity, and Verifiability.

Poor requirement: "The platform should make learning very easy."

Improved requirement: "The system shall display the learner's course progress and completed learning activities within 3 seconds under normal operating conditions."

The improved requirement is specific, measurable, and testable.

14. Assimilation into RE Model

Stakeholders → Requirement Elicitation → Gather Application Requirements → Requirement Analysis → Requirement Classification → Functional + Non-Functional Requirements → Requirement Specification → Requirement Validation → Validated Requirements → RE Model / SRS

The requirements gathered from different stakeholders are analyzed, classified, documented, and validated before being incorporated into the software development process.

15. Sample Use Cases

Use Case	Actor
Register Account	Learner / Teacher
Login	All Users
Browse Courses	Learner
Search Course	Learner
Enroll in Course	Learner
Access Learning Material	Learner
Attempt Quiz	Learner
Submit Assignment	Learner
Track Progress	Learner
Get Learning Recommendations	Learner
Manage Course	Teacher
Create Assessment	Teacher
View Learner Performance	Teacher
Manage Users and Platform	Administrator

16. Expected Outcome

After performing this practical, the student will be able to identify stakeholders, gather application-specific requirements, classify functional and non-functional requirements, prepare user and system requirements, document requirements in a structured format, and understand how gathered requirements are incorporated into the Requirement Engineering process.

17. Conclusion

The requirements for the Smart Education Platform were successfully gathered from different stakeholders using requirement elicitation techniques. The requirements were analyzed and classified into functional, non-functional, user, system, business, and security requirements. The validated requirements can be further used for preparing the Software Requirement Specification (SRS) and developing the Requirement Engineering model.